



Fig. 4: Author's prototype

begin(9600) function in the Arduino sketch.

GPS output. The GPS receiver output can be viewed on the serial monitor of the Arduino as shown in Fig. 3. In order to get this GPS data,

use the same connections as mentioned above but remove microcontroller ATmega328 from the Arduino board. Then open the Arduino IDE and select Serial Monitor option to view GPS output format.

This window shows many strings, of which you need to use only \$GPRMC. Here, after the first comma you can see time in 24-hour format, and after the ninth comma you can see date. Arduino program extracts date and time from this \$GPRMC string, processes them and displays on LCD1.

Software

Circuit operation is controlled by the software program loaded into the internal memory of Arduino Uno. The program (gpsindia.ino) is written in Arduino programming language. Arduino IDE 1.6.4 is used to compile and upload the program.

Connect Arduino board to the PC and select the correct COM port in Arduino IDE. Compile the program/sketch. Select the correct board from Tools→Board menu in Arduino IDE

EFY Note

The source code and other information relating to this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

and upload the sketch. In this project, external header files are not required for programming.

Construction and testing

The circuit need not be assembled on a PCB. Connect the circuit using external header jumpers between Arduino Uno, LCD1 and SIM28M GPS receiver module. Connect a 9V power supply adaptor each to the SIM28M GPS receiver module and Arduino. Also connect ANT.1 to GPS1 module. Place the unit with LCD1 at a suitable location.

The author's prototype is shown in Fig. 4. **EFY**



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